McMaster University ENGINEERING | mohawk | W BOOTH SCHOOL OF ENGINEERING PRACTICE AND TECHNOLOGY
McMaster-Mohawk Bachelor of Technology Partnership

Laboratory 4
Tilt Sensor

PROC TECH 3CC3

Cloud Computing and Internet of Things

B.Tech – PROCTECH

PRELAB

Please read the following material prior to this lab:

1. ***Read about LabVIEW Shift Registers in NI documentation:*** https://www.ni.com/docs/en-US/bundle/labview/page/lvconcepts/shift_registers_concepts.html

OBJECTIVES

In this lab you will learn the following:

1. Build a **digital circuit** using a Tilt Ball Switch sensor to detect tilt, and activate a buzzer when tilt is detected.
2. Build a **LabVIEW VI** to display the output from the tilt sensor.
3. Build a **LabVIEW VI** to display the number of tilts.

MATERIALS

1. 1x Tilt Ball Switch
2. 1x 10k Ohm Resistor
3. 1x Active Buzzer
4. Jumper Wires
5. Arduino Mega
6. Breadboard

PROCEDURE

Exercise 1. Build a circuit for the Tilt Sensor & Active Buzzer

In this exercise, you will build a simple digital circuit for using the Tilt Sensor and Active Buzzer.

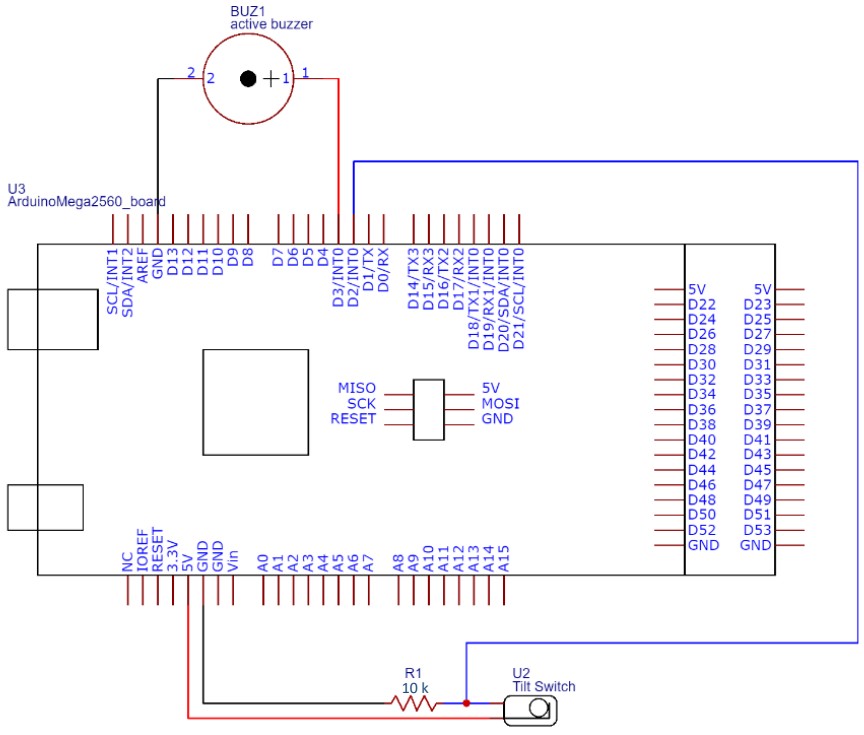
When the tilt sensor is upright, the sensor acts as a short circuit. When it is tilted, it acted as an open circuit. To prevent *floating* (When a digital read pin is connected to an open circuit, it has no reference ground to determine whether the signal is HIGH or LOW. This is referred to as *floating*) we will use a pull-down resistor configuration.

With a pull-down resistor configuration, when the sensor is short circuit (up-right), the digital read pin will read HIGH. When the sensor is open the pin will read LOW.

The active buzzer will be activated with a digital write pin. When the pin is sending a HIGH signal, the buzzer will turn on.

1. Wire your Tilt Ball Switch as shown below. You will be using Digital (2) Pin to read the sensor. Because the sensor is simply a switch, it does not matter which end you connect to the positive and negative terminals

2. Add the Active Buzzer. Simply attached positive terminal (marked by the + on the top of the buzzer) to the Digital (3) Pin, and the other terminal to **Ground (GND)**.



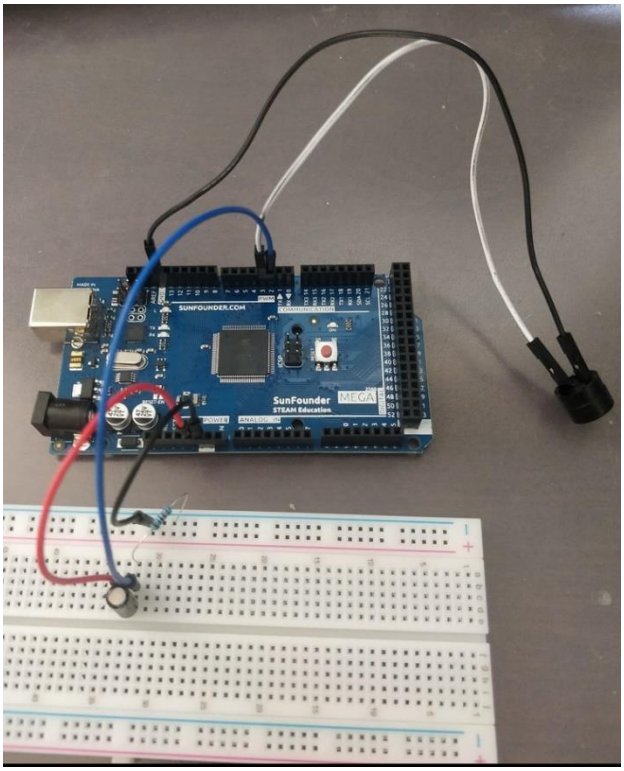


Figure 1: Complete Circuit

End of Exercise

Exercise 2. View Tilt Sensor output & Control Active Buzzer.

In this exercise, you will build a LabVIEW VI to sample and display the output from the Tilt sensor. When the sensor detects tilt, the VI will turn on the active buzzer.

Front Panel

1. Open a new blank ***Labview*** Project.

2. Add a **Round LED** from **Controls»Modern»Boolean.** Label it as `Tilted`.



Block Diagram

1. Click on **Windows»Show Block Diagram** to display your block diagram.

2. Add the **LINX Open** function from the **Functions»MakerHub»LINX** palette.
    a. Right click on the `Serial Port` node of the **LINX Open** function and select **Create constant.**
    b. Select your comm port from the drop down.

3. Add a **LINX Digital Read** function from the **Functions»MakerHub»LINX»Peripherals» Digital** palette.
    a. Right click on the `DI Channel` node of the **LINX Digital Read** function and select **Create constant.**
    b. Write `2` into the constant.

4. Add a **LINX Digital Write** function from the **Functions»MakerHub»LINX»Peripherals» Digital** palette.
    a. Right click on the `DO Channel` node of the **LINX Digital Write** function and select **Create constant.**
    b. Write `3` into the constant.

5. Add the **LINX Close** function from the **Functions»MakerHub»LINX** palette.

6. Add a **Not** function from **Functions»Programming»Boolean**.

7. Add a **While Loop** from **Functions»Programming»Structure.** Encompass the loop around everything except for the **LINX Open & Close** functions.
    a. Add a **stop** button to the **While Loop**.
    b. Add a **Wait Until Next ms Multiple**  inside the **While Loop** from **Functions»Programming»Timing.**
    c. Set the `millisecond multiple` node of the **Wait Until Next ms Multiple** function to a numeric constant `100`. This programs the While Loop to repeat every 100ms.

8. Wire the diagram as shown in **Figure 2**. The **Not** function is used to write the inverse of the read signal. Because our sensor will output a HIGH signal when upright and a LOW signal when tilted, we should be sending the opposite to the active buzzer.

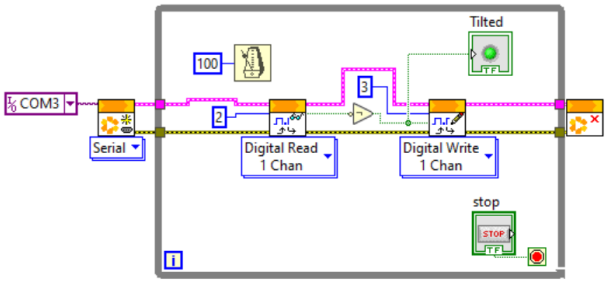


Figure 2: Complete Block Diagram

9. Run the program. Slowly tilt your sensor. Notice the LED light up & the buzzer activate when tilted.

End of Exercise

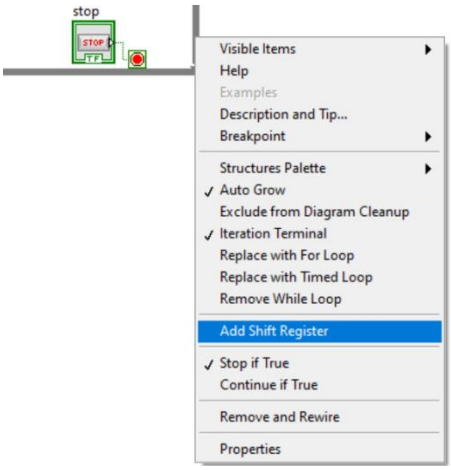
Exercise 3: Count the Number of Tilts

In this exercise, you will write a program to count and display the number of tilts that have been detected by the sensor. The counting will be done with the help of **shift registers**.

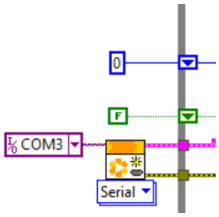
Block Diagram

1. Use the same VI and setup described from Exercise #2 (above).

2. Add two **Shift Registers** by Right clicking on the While Loop Border and selecting **Add Shift Register**. These registers are initialized at the start of the loop and **retain their values** each loop.



   a. Wire a **Numeric constant** from **Functions»Programming»Numeric** to one of the input (left side) shift register nodes. Set the value to 0. This register will store the tilt count.
   b. Wire a **False Constant** from **Functions»Programming»Boolean** to the other input (left side) shift register node. This register will store the digital read reading from the previous iteration of the loop.



3. Add a **Not Equal?** function from **Functions»Programming»Comparison.** This

4. Add a **And** function from **Functions»Programming»Boolean.**

5. Add a **Boolean To (1,0)** function from **Functions»Programming»Boolean.**

6. Add a **Add** function from **Functions»Programming»Numeric.**

7. Wire the block diagram as shown in **Figure 3**.

*The right-side shift register input nodes save the input values from the **current loop** to the **next loop**. Those same values become the outputs from the left-side shift registers. For more information refer to the reading from the pre-lab.*

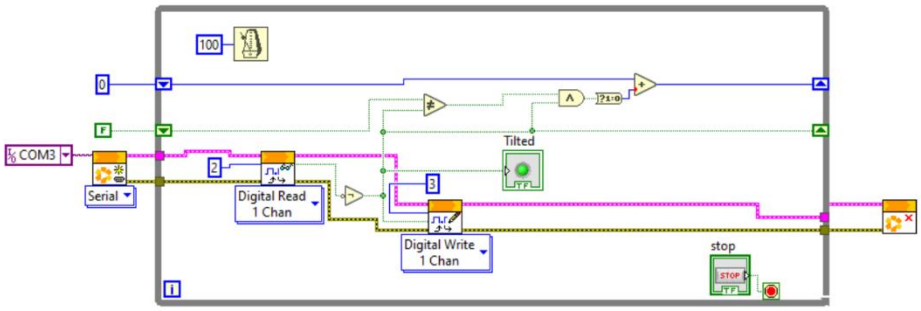


Figure 3: Block Diagram

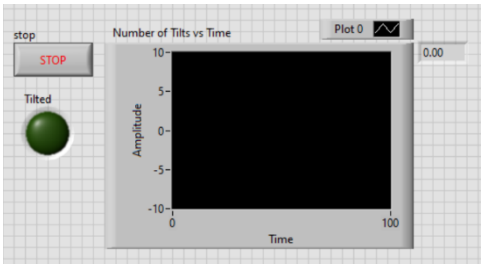
Front Panel

1. Go to the **front panel**.

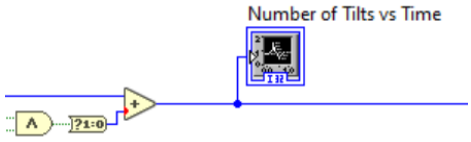
2. Add a **Waveform Chart** from the **Controls»Modern»Graph** palette.
    a. Right click on the **Waveform Chart** and enable the **digital display** from **Visible Items.**

3. Change the name Label to `Number of Tilts vs Time.`

4. Change the Y axis label to `Number of Tilts.`



5. Go back to the **block diagram** and wire the **Waveform Chart** to the output node of the **Add** function.



6. Click the **Run** button and verify that the Tilt Counter is working. Your counter should initialize to 0 and then increment by 1 every time a new tilt occurs.

7. Save this VI as lab4_ex3.vi.

End of Exercise

EXTEND YOUR KNOWLEDGE

1. Add a toggle switch control that will invert the behavior of the active buzzer. When toggled **ON**, the buzzer will activate when there is no tilt.

SUBMISSION

1. Use your own words to describe the operation of a tilt switch sensor and mention two of its applications.
2. Use your own words to describe how the sound is produced by an active buzzer and mention two of its applications.
3. State if the tilt switch and the active buzzer are input or output devices?
4. Explain the function of each of the two sets of shift registers in the VI and explain how the blocks in Figure 3 are used to count the Number of Tilts.
5. Include the new code corresponding to the extend your knowledge section and explain the program modification that you implemented.
6. Submit your completed VI file with your student names and numbers on it.